

Generative Scaffolding: A Framework for Ephemeral, Task-Scoped Interfaces

Final Proposal

Introduction

LLMs are constrained by static interfaces that negatively affect user context. Users need functional scaffolds that augment, rather than replace, active applications.

Research in computational notebooks show how ephemeral UIs can reduce the cognitive load of prompt engineering and understanding through direct manipulation and visual experimentation. However, a framework generalising these benefits beyond technical environments is absent.

This proposal introduces Ephemeral interfaces which are temporary, task-scoped components generated on demand. Design Science Research methodology will be used to develop a conceptual framework and a custom react library `react-ephemeral-experimental`. This artifact will serve to validate how transient, generative UIs impact task efficiency and user mental models in general productivity contexts.

Preliminary Literature Review

1. **Bieniek, Jan, Mohamed Rahouti, and Dinesh C. Verma.** "Generative AI in Multimodal User Interfaces: Trends, Challenges, and Cross-Platform Adaptability." *arXiv*, 15 Nov. 2024, <https://arxiv.org/pdf/2411.10234>.
2. **Leviathan, Yaniv, et al.** "Generative UI: LLMs are Effective UI Generators." *Google Research*, 18 Nov. 2024, <https://generativeui.github.io/static/pdfs/paper.pdf>.
3. **Sundar, Aparna, et al.** "The AI Interface: Designing for the Ideal Machine-Human Experience." *Computers in Human Behavior*, 2024, <https://www.sciencedirect.com/science/article/abs/pii/S0747563224004072>.
4. **"Multimodal Interface."** *ScienceDirect Topics*, Elsevier, <https://www.sciencedirect.com/topics/computer-science/multimodal-interface>. Accessed 26 Nov. 2025.
5. **Cheng, Ruijia, et al.** "BISCUIT: Scaffolding LLM-Generated Code with Ephemeral UIs in Computational Notebooks." *arXiv*, 10 Apr. 2024, <https://arxiv.org/pdf/2404.07387>.

Research Question or Objective

Primary Research Question:

How can ephemeral, task-scoped generative UI components be designed and implemented to augment existing applications without disrupting UX?

Supporting Questions:

1. What taxonomy of ephemeral interface patterns captures a range of situations?
2. What lifecycle model best describes the creation, activation, use, and dismissal of temporary generated components?
3. How do users develop mental models of UIs with dynamically appearing and disappearing elements?
4. What task types do ephemeral generated interfaces provide the greatest benefit over static alternatives?

Objectives:

- Make a conceptual framework for differentiating ephemeral interfaces from full page generative UI.
- Propose design principles for creating temporary components that integrate naturally within host applications.
- Implement a custom React library ([react-ephemeral-experimental](#)) as a research instrument to test the framework.

Execution Plan

Phase	Timeline	Activities	Deliverables
Literature Review	December	Initial research	Annotated bibliography, gap analysis
Framework Design	December	Develop taxonomy, lifecycle model, context protocol	Conceptual framework section
Instrument Dev	January	Build react library and Host App	NPM library
Main Study	February	Data collection with 30-40 participants using the Host App	Raw data
Analysis	February	Statistical analysis, qualitative coding	Results summary
Writing	May	Final paper, presentation preparation	Completed capstone

Desired Measurable Outcome

- Achieve more than 15% higher task completion rates compared to static or baseline interfaces.
- Demonstrate significantly lower intrusiveness scores of $p < 0.05$ than traditional modal dialogs.
- Secure more than 65% preference for ephemeral assistance over always-visible or traditional alternatives.